

Parameter Justification Reference — SAGE IRL Extension & SRL Conference Paper

A working reference for every hand-set constant in the system: the defensible rationale for each, which justification pattern it uses, which paper it belongs to, and ready-to-adapt paper text. Drop the pieces into the papers where they fit. Last updated: 2026-07-24.

Part 1 — The Five Justification Patterns

Nearly every constant in the system is defended by one (or more) of five recurring patterns. State these once in each paper (see the Parameter Selection paragraph, Part 4), then let the per-constant tables do the itemized work. Repetition across constants is a feature: it shows the choices follow a small number of principles rather than seventeen ad-hoc picks.

Pattern 1 — Policy threshold (consequence-stated, configurable). The value is not an empirical estimate but an institutional confidence bar, analogous to the $p < 0.05$ convention. It is justified by its *consequences* (which the evaluation reports), and it is instructor-configurable — which is thematically native to instructor-regulated learning. A reviewer can disagree with a policy; they cannot call it unjustified, because policy values are justified by their reported consequences.

Pattern 2 — Theory-driven ordering (literature fixes the direction, not the digit). The *relative ordering* of values across tiers follows established results (e.g., recognition decays slower than skill execution; guessing is easier on multiple choice than in working code). The specific digits are engineering defaults within literature-consistent ranges. Cite the ordering; sweep the digits.

Pattern 3 — Window logic (constants sized to "about three events"). EMA smoothing factors, trigger counts, and minimum-evidence requirements are all sized to the same behavioral window: one event is noise, two is coincidence, three is a pattern. Several

constants deliberately share this window so the system has one uniform "minimum evidence" convention.

Pattern 4 — Guarantee-invariance (the formal results do not touch it). The paper's theorems and propositions hold *for all admissible parameter values*. These constants affect only secondary quantities (time-to- certification, display, presentation) — never the truth of the guarantees. This is the strongest pattern; use it loudly wherever it applies.

Pattern 5 — Adaptivity (defaults that data overwrites). The constant is only an initialization; a deployed mechanism recalibrates it (SM-2 ease factors modify decay; Dirichlet counting recalibrates MCN CPTs; sensitivity sweeps bound the trajectory thresholds). Defaults matter only until data arrives.

The overarching stance (say this once, own it): for most constants there is no reason the value is 0.10 rather than 0.12, and no honest phrasing manufactures one. What exists — and what reviewers accept — is (i) a reason the value is in a particular *range*, and (ii) evidence the conclusions are *stable* across that range. "Why this value?" is answered by "any value here behaves the same; here is the sweep." The sensitivity sweeps are therefore load-bearing, not garnish: without them, stability is an assertion; with them, every row below closes.

Part 2 — Per-Constant Justifications

Grouped by subsystem. Columns: value · pattern(s) · the rationale · which paper.

2.1 Certification & BKT core (IRL extension)

Constant	Value	Pattern	Rationale	Paper
θ_{cert}	0.95	1	Policy confidence bar, analogous to the 95% convention; instructor-configurable. Consequence-justified: at 0.95 a true master certifies in a handful of clean demonstrations per tier — verified against the deployed update: quiz reaches 0.917 at n=3 and crosses 0.95 at n=4 , so state " ~ 4 " (or the exact recomputed per-tier counts), not " ~ 3 ". A lower bar certifies faster but admits more false positives — the evaluation measures exactly that trade-off.	IRL
θ_{dec}	0.75	1	Must sit well below θ_{cert} for Schmitt-trigger hysteresis to prevent oscillation at the boundary. The <i>gap</i> , not the value, is what matters.	IRL
N_{min}	3	3	Smallest count that distinguishes a pattern from a coincidence; also the minimum for the guess-run bound P_{G}^3 to be meaningfully small. $N=2$ admits lucky pairs; $N \geq 4$ only delays true masters. Shares the system-wide "three events" convention with N_{trigger} .	IRL
P_{G} (per tier)	0.20 / 0.10 / 0.05	2, 4	Ordering is structural: a 4-option MCQ has a ~ 0.25 blind-guess floor, so 0.20 sits just below chance; producing a correct line of code or a working program by luck is progressively rarer. Ranges consistent with the guess/slip bounds discussed by Baker et al. (2008). The digits are defaults within those ranges; the Evidence Diversity Guarantee is invariant to them (it holds for all admissible values), and their only effect — certification latency — is swept in the evaluation.	IRL

Constant	Value	Pattern	Rationale	Paper
P_S (per tier)	0.10 / 0.15 / 0.20	2, 4	Ordering is structural: more execution steps means more ways to slip despite knowledge (typos, off-by-one errors), so slip rises from declarative to applied. Same range-plus-sweep defense as P_G.	IRL
P_LO (per tier)	0.30 / 0.05 / 0.01	2, 5	Ordering: incoming students plausibly carry partial declarative exposure to C concepts, near-zero applied skill. Also a prior in the strict sense — washed out by evidence within a few observations, so the digit governs little.	IRL
P_T	0.09	2	Standard BKT transition magnitude (classical fits commonly land near 0.1). The <i>only-on-correct</i> application is a separate, deliberate design decision argued in the methodology (crediting learning after an incorrect attempt would let wrong answers drift the posterior upward).	IRL

2.2 Retention, load, misconceptions (IRL extension)

Constant	Value	Pattern	Rationale	Paper
Decay half-lives	14 / 7 / 5 days	2, 5	Ordering from the retention literature: recognition outlasts skill execution, so declarative decays slowest and applied fastest. The digits are defaults made <i>adaptive</i> by SM-2 ($\lambda \leftarrow \lambda/EF$ after successful review), so they govern only unpracticed material.	IRL
SM-2 ease floor	1.3	—	The literal SuperMemo SM-2 constant. Cite Woźniak. (The one externally sourced value — say so.)	IRL
Load weights	0.4 / 0.3 / 0.3	1	Near-uniform weighting with a mild lean toward the most direct signal (code complexity). Cognitive Load Theory motivates the <i>inputs</i> , not the weights. Crucially: the load estimate feeds presentation only — it never touches certification. State that explicitly; it converts a tuning question into a non-issue.	IRL
λ_M (misconception EMA)	0.7	3	EMA window $\approx$ last 3 errors dominate — the same "three events" window logic as N_{min} and $N_{trigger}$.	IRL
θ_M (misconception flag)	0.25	1	Consequence-stated: flags a persistent error pattern. [VERIFY BEFORE USE] The "cleared by two consecutive correct responses" rule was not confirmed in code (a <code>clear_concepts()</code> and streak-reset exist, but no explicit two-consecutive rule was found) — either locate the	IRL

Constant	Value	Pattern	Rationale	Paper
			rule or describe the actual clearing mechanism; do not state the two-consecutive claim on the current evidence.	
ATM/ adaptation cutoffs	0.75 / 0.35	1, 2	Kalyuga et al. (2003) establish the expertise- reversal <i>effect</i> , not thresholds — say so plainly. These cutoffs partition the display composite for scaffolding level only; they gate no formal quantity.	IRL

2.3 Calibration loop (SRL paper)

Constant	Value	Pattern	Rationale	Paper
α (blend)	0.6	1, 4	Majority weight to the evidence-based estimate, so self-report softens but cannot dominate the display. Any $\alpha > 0.5$ has this property; 0.6 is a modest choice within it. Certification-irrelevant by construction (P_{eff} is excluded from certification).	SRL
λ_{cal} (direction EMA)	0.4	3	EMA window $\approx$ last 3-4 adjustments — deliberately matched to N_{trigger} so the two trigger conditions measure the <i>same</i> behavioral window rather than two different ones.	SRL
N_{trigger}	3	3	One report is a mood, two is a coincidence, three is a pattern. Consistency with N_{min} gives the whole system a single uniform minimum-evidence convention.	SRL
D threshold	-0.3	3	Requires the majority of recent adjustments to point downward; filters mixed/oscillating signals. Any moderately negative value works; swept.	SRL
β (step)	0.10	1, 4	Sized so the cap is reached in 2-3 firings: no single report can saturate the adaptation, sustained reports do. The <i>cap</i> , not the step, carries the guarantee.	SRL
$P_{\text{G,max}}$ (per tier)	0.40 / 0.25 / 0.15	4 (derived)	The best-justified row: ceilings chosen strictly below the BKT non-degeneracy boundary $p_{\text{S}} + p_{\text{G}} < 1$ at every tier (tier sums 0.50 / 0.40 / 0.35), so a correct answer remains diagnostic even at full saturation. This is a <i>derivation from a model property</i> , not a pick	SRL

Constant	Value	Pattern	Rationale	Paper
			— and it is what makes clause (ii) of the no-self-certification proposition provable. Present it as such.	

2.4 Behavioral / metacognitive indicator cutoffs (SRL paper)

Constant	Value	Pattern	Rationale	Paper
Rushing cutoff	4 s ($\wedge$ mastery < 0.4)	1	Face-valid: 4 s is below plausible read-and-comprehend time for a question. A descriptive indicator surfaced to humans; gates nothing formal.	SRL
Impulsive give-up	< 5 s	1	Same face-validity logic; a skip faster than reading time is surrender, not consideration. Descriptive only. [VERIFY BEFORE USE] Code inspection found surrender detection to be keyword-based, with no confirmed 5-second timer — either locate the timer or describe the detection as implemented (keyword-triggered) and drop the "5 s" figure.	SRL
jol_cal, grit, etc.	various	1	The metacognition/calibration literature is qualitative here; these are descriptive indicators for the learner model and dashboard, not gating quantities. Say so once, collectively.	SRL

2.5 MCN conditional probability tables (SRL paper) — the honestly weak row

Constant	Value	Pattern	Rationale	Paper
MCN CPTs	hand- specified	5 (+ simulation)	The one place where hand-set numbers sit <i>inside</i> the inference itself, so a range-rationale alone does not cover it. The defense is two-part and must both appear: (i) the identifiability simulation showing the network recovers calibration states from evidence under the hand-set priors (mechanism works in principle); (ii) the Dirichlet counting path that recalibrates the CPTs from classroom co-occurrence data post-deployment. Present the CPTs as <i>priors with a stated update path</i> , not as fitted quantities. Do not paper over this row.	SRL

2.6 Security & trajectory constants (IRL extension)

Constant	Value	Pattern	Rationale	Paper
τ_{block} (hard block)	0.85	1	High bar because a false block carries direct pedagogical cost in a tutor (an axis the education setting adds over generic LLM security). Consequence-framed policy.	IRL
Mastery-widening slope (α)	0.05	1	Small by design: the security sandbox widens slowly with certified mastery, so earned competence buys latitude gradually rather than abruptly.	IRL
Strike / length / drift cutoffs	3 / 15 / 0.1 / 5	1, 3	Strike count uses the same "three events" window; the rest are face-valid operational cutoffs. Defend collectively, briefly.	IRL
TRAJ $_{\lambda}$ (accumulator decay)	0.8	1, 3	Behavioral meaning: memory half-life ≈ 3 turns; sustained risk saturates at $\sim 5\times$ per-turn risk. Matches the pacing of crescendo-style attacks, which assemble over 3–6 turns.	IRL
TRAJ τ_{judge}	0.45	1	Escalate before risk is even moderate — the conversation-level judge is cheap relative to a missed multi-turn assembly. Note (post-verification): benign sessions can and do cross this threshold (debugging reaches 0.49); that is by design and harmless, since crossing it consults the judge rather than blocking. τ_{judge} trades judge-invocation cost against escalation recall; it is not a benign/attack boundary and must not be presented as one.	IRL

Constant	Value	Pattern	Rationale	Paper
TRAJ τ_{block}	0.85	1	[INCONSISTENCY — resolve before deployment.] As deployed, risk ≥ 0.85 hard-blocks <i>before</i> the judge runs — a score-only correctness boundary, which contradicts the two-stage framing below, and the scores are known not to separate (benign systems-programming already reaches 0.72, leaving a 0.13 margin). A denser legitimate systems session could be hard-blocked with no judge in the loop. Two consistent resolutions: (a) recommended — route the $\geq \tau_{\text{block}}$ case through the judge as well (a mandatory, possibly stricter judge call rather than a bypass), so <i>every</i> block above τ_{judge} is judge-mediated and the "cost boundary, never a correctness boundary" claim holds without exception; the forgone LLM-call saving is negligible since ≥ 0.85 sessions are rare, and the paper's own false-block-cost principle argues for a judge exactly there. (b) fallback — keep the bypass as an explicit fail-safe, in which case the paper must state the exception honestly ("blocks above 0.85 are score-only") and report the observed benign-to-threshold margin (max benign 0.72), rather than claiming no benign session can be blocked on score alone. Do not ship the unqualified two-stage claim with the bypass in place.	IRL

Constant	Value	Pattern	Rationale	Paper
Peak/acc weights	0.5 / 0.5	1	Deliberately equal: a single spike and a slow burn are both concerning, and there is no principled reason to prefer either. Stating the equality as deliberate is the defense.	IRL
Trigger/classifier evidence (all trajectory τ)	—	—	[CORRECTED after code verification — do not use a separation argument.] The measured risk scores do <i>not</i> separate: attacks peak at 0.60–0.80, but benign debugging reaches 0.49 and benign systems-programming (fork/exec) 0.72 — inside the attack range. A separation plot would refute, not support, the thresholds. The honest and stronger defense is the two-stage framing: the <i>accumulator is a high-recall trigger</i> (verified: 3/3 attacks escalate to the judge across the entire $\lambda \in [0.7, 0.9] \times \tau_{\text{judge}} \in [0.35, 0.55]$ sweep), and the <i>transcript-fed judge is the classifier</i> (verified: 4/4 benign sessions — including the 0.49 debugging and 0.72 systems-programming sessions that escalate — are read in context and allowed). Consequently τ_{judge} is a cost boundary, not a correctness boundary: moving it changes how often the judge is consulted, never whether a benign student is blocked. Scope caveat: this claim currently holds for τ_{judge} only — as deployed, $\tau_{\text{block}} = 0.85$ hard-blocks <i>without</i> the judge, an exception that must be resolved (see the τ_{block} row above) before the unqualified two-stage claim can appear in the paper. Report the	IRL

Constant	Value	Pattern	Rationale	Paper
			two stages' results separately (escalation recall; post-judge false-block rate) instead of a score-distribution plot.	

2.7 Do not defend in the papers

Classifier margins (0.05), fuzzy-match threshold (0.86), rate limits, top_k, temperatures, message-size caps, and similar infrastructure conventions. Auditing them was diligent; justifying them in a paper would signal insecurity. No reviewer asks why top_k is what it is. Omit.

Part 3 — Which Sweeps Are Load-Bearing

The range-level rationales above hold only if the sweeps exist. One synthetic- learner harness serves all of them:

1. **(P_G, P_S) grid** → false-certification rate and time-to-certification are stable across the plausible region; the N_min requirement (hence the Guarantee) holds everywhere. (*IRL §IV — parameter robustness*)
2. **(β, λ_cal, α) sweep** → calibration-loop behavior (over-crediting reduction, no harm to well-calibrated learners) is stable. (*SRL §IV — simulation support*)
3. **(TRAJ_λ, τ_judge) sweep + two-stage results** → across the interval, escalation recall stays 3/3 (no attack evades the trigger) and the post-judge false-block rate stays 0/4 (no benign session is blocked, even those whose accumulator scores enter the attack range). Do **not** produce a score-separation plot — verification against the deployed scorer shows the distributions overlap (benign systems-programming reaches 0.72), and the plot would refute rather than support the thresholds. The claim the sweep defends is stability of the *two-stage outcome*, not separability of the scores. (*IRL §IV — adversarial robustness*)
4. **Bounded-delay recomputation** → regenerate the Corollary numbers from the exact deployed update; do not ship hand-traced sequences. This is now known to

be necessary, not precautionary: verification already shows the hand-traced baseline is off by one for the quiz tier (certification at $n=4$, not $n=3$), so both the baseline and the at-cap counts — and therefore the Corollary's "+2 per tier" claim — must come from the script. (*SRL §III — Corollary; also the θ_{cert} consequence claim in the IRL paper*)

Part 4 — Drop-In "Parameter Selection" Paragraphs

For the IRL extension (place at the end of §III or the start of §IV)

Parameter selection. The constants of the learner model fall into three kinds, and we distinguish them explicitly. First, *policy thresholds* ($\theta_{cert} = 0.95$, $N_{min} = 3$, and the decertification bound $\theta_{dec} = 0.75$) encode the institution's required confidence before unlocking downstream content; they are instructor-configurable quantities analogous to a confidence-level convention, justified by their measured consequences rather than fitted from data. Second, *model parameters* (the per-tier guess, slip, prior, and transition rates) are engineering defaults whose per-tier orderings follow established results — guessing is easier on recognition items than in working code, and recognition outlasts skill execution — and whose specific values affect only certification latency: the Evidence Diversity Guarantee of Section [X] holds for all admissible parameter values, and Section [Y] reports sensitivity of certification behavior across the plausible parameter region. Third, *display and presentation constants* (the composite ceilings and adaptation cutoffs) shape only what is shown to the learner and the scaffolding level; they are excluded from certification by construction.

For the SRL paper (place at the end of §III)

Parameter selection. The calibration loop's constants are sized to a single behavioral window: the trigger count ($N = 3$), the direction average ($\lambda_{cal} = 0.4$, weighting roughly the last three to four adjustments), and the sustained-direction threshold ($D < -0.3$) all operationalize the same requirement — that one self-report is a mood, two a coincidence, and three a pattern — mirroring the minimum-evidence convention of the underlying cognitive model. The adaptation step $\beta = 0.10$ is sized so that no single

report can saturate the adaptation. Crucially, the formal properties of Section [X] are invariant to all of these choices: the no-self-certification guarantee requires only that adaptation be monotone non-decreasing up to the per-tier ceilings, and the ceilings themselves are not free parameters but are derived from the BKT non-degeneracy condition $p_S + p_G < 1$, which is what makes clause (ii) provable. Section [Y] reports sensitivity of the loop's behavior across the plausible range of the remaining constants.

Part 5 — Language Audit Checklist

Before submission, sweep both drafts for overselling verbs. Reviewers forgive defaults; they do not forgive defaults dressed as fits.

- Search for: "calibrated", "optimized", "tuned", "chosen to", "fitted", "learned" — anywhere these describe a hand-set constant, replace with "set", "a default of", or "a policy of".
 - Every parameter table gets a *Rationale* column (or caption sentence) carrying the one-line justification from Part 2.
 - The MCN CPTs must never be described as anything other than hand-specified priors with a stated recalibration path.
 - The SM-2 floor (1.3) is the one value with a literal external source — cite it, and let it quietly demonstrate that you cite sources where they exist.
 - **Every empirical-sounding number in a rationale must be recomputed from the deployed code before it ships** — the "~3 demonstrations" claim and the score-separation claim in earlier versions of this document were both contradicted by the code. A rationale that recommends a plot must survive the plot actually being made.
 - For the trajectory thresholds specifically: never use "separation" language. The defense is trigger recall + judge precision, reported as two stages.
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Part 6 — Code-Verification Status (2026-07-24 audit)

Verified and safe to lean on hard:

- $P_{G,max}$ derivation from $p_S + p_G < 1$ (sums 0.50/0.40/0.35) — the document's strongest row, confirmed by computation.
- P_T only-on-correct as deliberate design — confirmed verbatim in code comments.
- All BKT digits, thresholds, and the effective $D = -0.3$ — match deployed code.
- Load-never-touches-certification and ATM-gates-nothing-formal — confirmed by architecture (presentation/Sentinel paths only).
- Trigger recall 3/3 across the full λ/τ sweep — confirmed.

Corrected in this revision:

- Score-separation claim (§2.6, Part 3 item 3) — false; replaced with the trigger/classifier framing.
- θ_{cert} "~3 demonstrations" — off by one ($n=4$ for quiz); recompute per tier.

Flagged, pending verification:

- Misconception two-consecutive-correct clearing rule (§2.2).
- Impulsive give-up 5-second timer (§2.4).

Pending design decision (pre-deployment, latent not observed):

- $\tau_{block} = 0.85$ bypasses the judge (code returns before the judge runs), contradicting the two-stage framing given non-separating scores (max benign observed 0.72; margin 0.13). Resolve per the τ_{block} row: route the hard block through the judge (recommended) or keep the bypass and state the exception explicitly in the paper. Live tests did not trip it — this is the same class of overclaim corrected elsewhere in this revision, caught before it shipped.